[Q1] Simplify $(2\times10^5) \times (3\times10^{-2})$ and express the answer in scientific notation.

A. 6×10^{-3}

B. 6×10^7

C. 6×10^3

D. 5×10^3

[Q2] Factorize $x^2 - 8x + 16 - y^2$.

A. $(x-4-y)(x-4+y)$

B. $(x-4)(x-4-y)$

C. $(x+y-4)(x-y+4)$

D. $(x-y-4)(x-y+4)$

[Q3] The polynomial $f(x) = 2x^3 - 5x^2 + ax + 6$ is exactly divisible by $(x - 2)$. Find the value of a.

A. 1

B. 5

C. -3

D. -1

[Q4] If the equation $2x^2 + (m-1)x + 8 = 0$ has no real roots, find the range of values of m.

A. $m < -7$ or $m > 9$

B. $-7 < m < 9$

C. $-9 < m < 7$

D. $1 < m < 8$

[Q5] w varies directly as u and inversely as v^2. If u increases by 50% and v decreases by 20%, find the percentage change in w.

A. increase by 134.375%

B. increase by 50%

C. increase by 87.5%

D. decrease by 20%

[Q6] Three numbers 2, x, 18 form a geometric sequence in this order. Find the value of x (x is positive).

A. 10
B. 8
C. 6
D. 36

[Q7] Solve the equation $\log_2(x - 1) + \log_2(x - 3) = 3$.

A. -1
B. 5
C. 3
D. 11

[Q8] The parabola $y = ax^2 + bx + c$ has vertex $(2, -5)$ and passes through $(0, 3)$. Find the value of a.

A. 1
B. 8
C. -2
D. 2

[Q9] Find the number of integers x satisfying $-6 \le x$, $-2x + 5 \ge -9$ and $3x - 4 < 11$.

A. 13
B. 12
C. 10
D. 11

[Q10] Using the graphs of $y = x^2 - 4$ and $y = 2x$, determine the number of roots of the equation $x^2 - 2x - 4 = 0$.

A. 0
B. 2
C. 1
D. Cannot be determined

[Q11] Line L_1: $2x + 3y - 6 = 0$ is parallel to line L_2: $4x + ky + 5 = 0$. Find the value of k.

A. 6

B. $\frac{3}{2}$

C. -6

D. $\frac{8}{3}$

[Q12] The circle C: $x^2 + y^2 + 4x - 2y - 20 = 0$. Determine the position of P(3, 1) relative to circle C.

A. P lies inside the circle

B. P lies outside the circle

C. P lies on the circle

D. Cannot be determined

[Q13] Refer to the image. A, B, C, D are points on a circle, AC is a diameter, and $\angle BAC = 28°$. Find $\angle BDC$.

A. 62°

B. 28°

C. 56°

D. 90°

[Q14] In $\triangle ABC$, $a = 7$, $b = 8$, $c = 13$. Find the size of $\angle C$.

A. 60°

B. 90°

C. 150°

D. 120°

[Q15] Find the solutions of $2\cos\theta + 1 = 0$ in the range $0° \leq \theta < 360°$.

A. 60°, 300°

B. 150°, 210°

C. 120°, 240°

D. 60°, 240°

[Q16] 3 different gifts are to be distributed to A, B and C (one each) chosen from 8 distinct gift types. Find the number of ways.

A. 336

B. 56

C. 512

D. 24

[Q17] A bag has 20 cards numbered 1 to 20. One card is drawn at random. Find the probability that the number is a multiple of 3 or a multiple of 5.

A. $\frac{9}{20}$

B. $\frac{11}{20}$

C. $\frac{2}{5}$

D. $\frac{1}{2}$

[Q18] A stem-and-leaf diagram of a data set shows median 34 and mode 28. If a new value 50 is added, which of the following statistics must change?

A. Mode only

B. Median only

C. Range

D.
Mode and Median both remain unchanged

[Q19] A set of 5 data: 3, 5, 7, 9, 11 has standard deviation σ. Find the variance of this data set.

A. $\sqrt{8}$

B. 8

C. $2\sqrt{2}$

D. 4

[Q20] A moving point P(x, y) is always 2 units farther from the fixed point F(0, 2) than from the x-axis. Find the equation of the locus of P.

A. $x^2 = -8y$

B. $x^2 = 4y$

C. $y^2 = 8x$

D. $x^2 = 8y$

[Q21] Given $f(x) = 2x - 3$ and $g(x) = x^2 + 1$, find the value of $f(g(2))$.

A. 11

B. 5

C. 9

D. 7

[Q22] The polynomial equation $2x^3 - 3x^2 - 11x + 6 = 0$ has one root equal to 3. Find the sum of the other two roots.

A. -1.5

B. 0

C. 1.5

D. -3

[Q23] Find the value(s) of k such that the line $y = x + k$ is tangent to the circle $x^2 + y^2 = 8$.

A. $k = \pm 2$

B. $k = \pm 2\sqrt{2}$

C. $k = \pm 4$

D. $k = \pm\sqrt{8}$

[Q24] A city has a current population of 500,000, growing at 4% per year. Find approximately how many years are needed for the population to exceed 800,000 ($\log 2 = 0.301$, $\log 1.04 = 0.017$).

A. 8

B. 12

C. 10

D. 14

[Q25] Bag A has 4 white and 6 black balls; Bag B has 5 white and 3 black balls. A bag is chosen at random, then a ball is drawn from it. Find the probability of drawing a white ball.

A. $\frac{9}{16}$

B. $\frac{9}{20}$

C. $\frac{41}{80}$

D. $\frac{1}{2}$

[Q26] Subject to $x \geq 0$, $y \geq 0$, $x + 2y \leq 10$, $3x + y \leq 15$, find the maximum value of $P = 4x + 3y$.

A. 20

B. 25

C. 30

D. 24

[Q27] Refer to the image. ABCD is a cyclic quadrilateral with $\angle A : \angle C = 2 : 3$. Find the size of $\angle C$.

A. 72°

B. 60°

C. 120°

D. 108°

[Q28] The graph of $y = f(x)$ is first vertically stretched by a scale factor of 2, then translated 3 units to the left. If $f(x) = x^2 - 1$, find the equation of the resulting graph.

A. $y = 2(x + 3)^2 - 2$

B. $y = 2(x + 3)^2 - 1$

C. $y = 2x^2 + 12x + 16$

D. $y = 2x^2 - 1$

[Q29] Solve the simultaneous equations: $y = x^2 - 2x$, $y = 3x - 4$.

A. $x = -1, 4$

B. $x = 1, 4$

C. $x = 1, -4$

D. $x = 2, -1$

[Q30] Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

A. $3x + 4y = 7$

B. $4x + 3y = 25$

C. $3x - 4y = 25$

D. $3x + 4y = 25$

[Q31] Refer to the image. Two chords PQ and RS of a circle intersect at point T inside the circle. PT = 6, TQ = 8, RT = 4. Find the length TS, and verify that T must lie inside the circle.

A. 12

B. 10

C. 8

D. 6

[Q32] Given that x, y are positive numbers with $2^x = 6$ and $3^y = 6$, find the value of 1/x + 1/y.

A. $\frac{1}{\log 6}$

B. $\log 6$

C. 1

D. 2

[Q33] In △ABC, ∠A = 60°, b = 6, c = 8. Find the area of △ABC.

A. 12

B. 24

C. $24\sqrt{3}$

D. $12\sqrt{3}$

[Q34] Refer to the image. In a rectangular box ABCDA'B'C'D', AB = 6, BC = 8, AA' = 10. Find the shortest path along the surface of the box from A to C' (using surface unfolding).

A. $10\sqrt{2}$

B. $\sqrt{356}$

C. $2\sqrt{74}$

D. 18

[Q35] A fair die is tossed 4 times. Find the probability of getting at least one 6.

A. $1 - \left(\frac{5}{6}\right)^4$

B. $\frac{4}{6}$

C. $\left(\frac{1}{6}\right)^4$

D. $\frac{1}{6} \times 4$

[Q36] Refer to the image. The circle has equation $x^2 + y^2 - 2x + 4y - 20 = 0$. Find the length of the chord cut by the line $y = 2$.

A. 8

B. 10

C. 6

D. $4\sqrt{5}$

[Q37] A factory produces a component with a fixed cost of $5000 and a variable cost of 20$ per unit. Each unit sells for $35. Let x be the number of units produced and sold. Find the smallest integer value of x for the factory to start making a profit (total revenue exceeds total cost).

A. 334

B. 333

C. 143

D. 250

[Q38] A moving point P satisfies the condition that the product of its distances from the lines $x = 0$ and $y = 0$ is always 9. Find the minimum distance from the origin to a point on this locus.

A. 3

B. 9

C. 6

D. $3\sqrt{2}$

[Q39] A farmer has 200 metres of fencing to enclose a rectangular area using an existing straight wall as one side (the wall side requires no fencing). Find the maximum area that can be enclosed.

A. 10000 m^2

B. 5000 m^2

C. 2500 m^2

D. 20000 m^2

[Q40] Given circles C_1: $x^2 + y^2 = 4$ and C_2: $(x - 6)^2 + y^2 = 4$, a moving circle M is externally tangent to both C_1 and C_2. Find the equation of the locus of the centre of M and describe its geometric shape.

A. $\frac{x^2}{9} + \frac{y^2}{5} = 1$ (an ellipse)

B. $x^2 + y^2 = 9$ (a circle)

C. $x = 3$ (a straight line, the perpendicular bisector of C_1C_2)

D. $y^2 = 12(x - 3)$ (a parabola)

[Q41] A, B and C independently attempt to solve a difficult problem, with probabilities of success 0.6, 0.5 and 0.4 respectively. Find the probability that exactly two of them succeed.

A. 0.12

B. 0.44

C. 0.38

D. 0.62

[Q42] A regular tetrahedron has edge length 6. Find the radius of its inscribed sphere (tangent to all four faces). (Hint: the height of the tetrahedron is $h = (6\sqrt{6})/3$.)

A. $\sqrt{6}$

B. $\frac{\sqrt{6}}{2}$

C. $\frac{3\sqrt{6}}{2}$

D. $2\sqrt{6}$

[Q43] Let x, y be real numbers satisfying $x^2 + y^2 \leq 25$ and $x + y \geq 5$. Find the maximum value of $z = 3x + 4y$.

A. 20

B. 35

C. 15

D. 25

[Q44] Refer to the image. A pyramid has a square base ABCD with side 10, with four congruent isosceles triangular faces. The dihedral angle between a slant face and the base is 60°. Find the volume of the pyramid.

A. $\frac{500\sqrt{3}}{3}$

B. $500\sqrt{3}$

C. $\frac{250\sqrt{3}}{3}$

D. 1000

[Q45] An arithmetic sequence $\{a_n\}$ has first term $a_1 = 20$ and common difference $d = -3$. Let S_n be the sum of the first n terms. Find the value of n that maximizes S_n, and find this maximum value.

A. $n = 8,\ S_n = 76$

B. $n = 7,\ S_n = 77$

C. $n = 7,\ S_n = 80$

D. $n = 6,\ S_n = 75$